

Network_Task_02_Anju

Research Network Devices

1. Router

Purpose: Connects different networks and provides internet access.

How it works: Receives data packets and forwards them to the correct destination.

Real-world usage: Home WiFi routers.

2. Switch

Purpose: Connects devices within a LAN.

How it works: Sends data only to the intended device using MAC addresses.

Real-world usage: Office networks.

3. Hub

Purpose: Connects multiple devices.

How it works: Broadcasts data to all connected devices.

Real-world usage: Small or old networks.

4. Access Point

Purpose: Provides wireless connectivity.

How it works: Extends a wired network to WiFi devices.

Real-world usage: College and office WiFi.

5. Firewall

Purpose: Protects the network from unauthorized access.

How it works: Filters incoming and outgoing traffic.

Real-world usage: Windows Firewall.

6. Modem

Purpose: Connects your home network to your ISP.

How it works: Converts ISP signals into internet access.

Real-world usage: JioFiber modem.

IP Address Classification

IP Address	Type	Reason
192.168.1.10	Private	Falls in 192.168.x.x private range
10.0.0.5	Private	Falls in 10.x.x.x private range
172.16.5.20	Private	Falls in 172.16–172.31 private range
8.8.8.8	Public	Google's public DNS
1.1.1.1	Public	Cloudflare public DNS
192.168.100.1	Private	Falls in 192.168.x.x private range

Find Network Information

ipconfig /all

Item	Value
IPv4 Address	192.168.**.**
Default Gateway	***.***.31.*
DNS Server	***.***.31.

Answer Network Questions

1. Which IP range does your device belong to?

192.168.**.** belongs to the 192.168.x.x private IP range.

2. Is it Public or Private?

Private IP Address.

3. What role does your router play?

The router connects my local network to the internet and forwards data between devices and external networks.

4. What would happen if DNS stopped working?

Websites could not be accessed using names like google.com because domain names would not be translated into IP addresses.

Network Communication Flow Diagram

YOUR DEVICE



JIOFIBER ROUTER



DNS SERVER



GOOGLE SERVER



RESPONSE BACK



YOUR DEVICE

Explain the Flow

➤ Step 1

The user enters www.google.com in the browser.

➤ Step 2

The request reaches the router and is forwarded to a DNS server.

➤ Step 3

The DNS server translates google.com into an IP address.

➤ Step 4

The request is sent to Google's server.

➤ Step 5

Google processes the request and sends data back.

➤ Step 6

The browser displays the webpage.

Practical Commands

1. What IP address did DNS return for Google?

142.***.**.***

2. Was the ping successful?

YES, THE PING WAS SUCCESSFUL BECAUSE ALL PACKETS WERE RECEIVED WITHOUT LOSS.

2. WHY IS DNS IMPORTANT BEFORE COMMUNICATION BEGINS?

DNS IS IMPORTANT BECAUSE IT CONVERTS HUMAN-READABLE DOMAIN NAMES SUCH AS GOOGLE.COM INTO IP ADDRESSES THAT COMPUTERS CAN USE FOR COMMUNICATION.

```
C:\Users\Anju sahana L>ping google.com

Pinging google.com [2404:6800:4009:824::200e] with 32 bytes of data:
Reply from 2404:6800:4009:824::200e: time=37ms
Reply from 2404:6800:4009:824::200e: time=32ms
Reply from 2404:6800:4009:824::200e: time=69ms
Reply from 2404:6800:4009:824::200e: time=63ms

Ping statistics for 2404:6800:4009:824::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 32ms, Maximum = 69ms, Average = 50ms
```